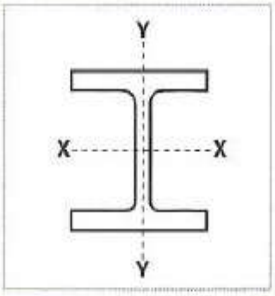


4a (i) Beams (Imperial Units) - Part I

	Designation Size		Unit Weight		Height Of Beam	Width Of Beam	Thickness		Root Radius	Depth Between Fillets	Area Of Section	Ratios For Local Buckling		Dimensions For Detailing		
							Flange	Web				Flange	Web	End Clearance	Notch	
					H	B	T	t	r	d	A	b/T	d/t	C	N	n
	in	(mm)	lb/ft	kg/m	mm	mm	mm	mm	mm	mm	cm²			mm	mm	mm
W33	33 x 11 1/2	(838 x 292)	118	175.6	834.9	291.7	18.8	14.0	17.8	761.7	224	7.76	54.4	9	149	37
			130	193.5	840.7	292.4	21.7	14.7	17.8	761.7	247	6.74	51.8	9	149	40
			141	209.8	845.8	293.0	24.4	15.4	17.8	761.4	268	6.01	49.5	10	149	42
			152	226.2	850.9	293.8	26.8	16.1	17.8	761.7	289	5.48	47.3	10	149	45
			169	251.5	859.0	292.1	31.0	17.0	17.8	761.4	319	4.71	44.8	11	148	49
			187	278.3	865.1	294.1	34.0	19.1	17.8	761.5	355	4.33	39.9	12	148	52
			204	303.6	871.2	295.7	37.1	20.6	17.8	761.4	386	3.99	37.0	12	148	55
			219	325.9	877.1	297.2	39.9	22.1	17.8	761.7	416	3.72	34.5	13	148	58
			243	361.6	885.2	299.7	43.9	24.4	17.8	761.8	460	3.41	31.2	14	148	62
			271	403.3	895.1	302.1	49.0	26.9	17.8	761.5	513	3.08	28.3	15	148	67
			301	447.9	905.0	305.2	54.1	30.0	17.8	761.2	572	2.82	25.4	17	148	72
			332	494.1	915.2	308.1	58.9	33.0	17.8	761.8	629	2.62	23.1	19	148	77
			361	537.2	925.1	310.6	64.0	35.6	17.8	761.5	684	2.43	21.4	20	148	82
	33 x 15 3/4	(838 x 400)	201	299.1	855.5	399.9	29.2	18.2	17.8	761.5	381	6.85	41.9	11	201	47
			221	328.9	861.8	401.4	32.4	19.7	17.8	761.4	420	6.20	38.7	12	201	50
			241	358.6	868.2	402.8	35.6	21.1	17.8	761.5	457	5.66	36.1	13	201	53
			263	391.4	877.1	401.4	39.9	22.1	17.8	761.7	499	5.03	34.5	13	200	58
			291	433.1	884.9	404.0	43.9	24.4	17.8	761.4	552	4.60	31.2	14	200	62
			318	473.2	893.1	406.0	48.0	26.4	17.8	761.5	603	4.23	28.8	15	200	66
			354	526.8	903.0	408.9	53.1	29.5	17.8	761.2	672	3.85	25.8	17	200	71
			387	575.9	913.1	411.5	57.9	32.0	17.8	761.7	734	3.55	23.8	18	200	76
			424	631.0	923.0	414.0	63.0	35.1	20.0	757.0	805	3.29	21.6	20	199	83
			468	696.5	935.0	418.0	69.1	38.6	20.0	756.8	889	3.02	19.6	21	200	89
W36	36 x 12	(914 x 305)	135	200.9	903.0	303.3	20.2	15.1	19.1	824.4	256	7.51	54.6	10	154	39
			150	223.2	910.4	304.1	23.9	15.9	19.1	824.4	286	6.36	51.8	10	154	43
			160	238.1	914.7	304.8	25.9	16.5	19.0	824.9	304	5.88	50.0	10	154	45
			170	253.0	918.4	305.5	27.9	17.3	19.1	824.4	323	5.47	47.7	11	154	47
			182	270.8	922.8	306.7	30.0	18.4	19.0	824.9	346	5.12	44.8	11	154	49
			194	288.7	926.6	307.7	32.0	19.5	19.1	824.4	368	4.81	42.3	12	154	51
			210	312.5	931.9	309.4	34.5	21.1	19.0	824.8	399	4.48	39.1	13	154	54
			232	345.3	942.9	307.8	39.9	22.1	19.1	824.9	440	3.86	37.3	13	153	59
			256	381.0	950.7	310.3	43.9	24.4	19.1	824.7	486	3.53	33.8	14	153	63
			286	425.6	960.9	312.8	48.0	26.9	19.1	826.7	536	3.26	30.7	15	153	67
			318	473.2	970.8	315.7	54.1	29.9	19.1	824.4	603	2.92	27.6	17	153	73
			350	520.9	980.7	318.8	58.9	33.0	19.1	824.7	663	2.71	25.0	19	153	78
			387	575.9	992.9	321.8	65.0	36.1	19.1	824.7	733	2.48	22.8	20	153	84
	36 x 16 1/2	(914 x 419)	230	342.3	911.8	418.5	32.0	19.4	24.1	799.6	437	6.54	41.2	12	210	56
			245	364.6	916.4	419.4	34.3	20.3	24.1	799.6	465	6.12	39.4	12	210	58
			260	386.9	921.0	420.5	36.6	21.4	24.1	799.6	494	5.74	37.4	13	210	61
			280	416.7	927.6	421.5	39.9	22.5	24.1	799.6	532	5.28	35.6	13	210	64
			300	446.4	933.2	423.0	42.7	24.0	24.1	799.7	569	4.96	33.3	14	210	67
			328	488.1	942.1	422.4	47.0	25.9	24.1	799.9	622	4.49	30.9	15	208	71
			359	534.2	950.0	424.9	51.1	28.5	24.1	799.7	680	4.16	28.1	16	208	75
			393	584.8	960.1	427.5	55.9	31.0	24.1	800.1	746	3.83	25.8	17	208	80
			439	653.3	972.0	431.0	62.0	34.5	24.0	800.0	832	3.48	23.2	19	208	86
			527	784.3	996.0	437.0	73.9	40.9	24.0	800.2	998	2.96	19.6	22	208	98
			650	967.3	1028.0	446.0	89.9	50.0	24.0	800.2	1232	2.48	16.0	27	208	114

Note: Calculated based on 1 kg = 2.2046 lb and 1 m = 3.2808 feet.
*Please contact us for further clarification.



4a (i) Beams (Imperial Units) - Part II

Unit Weight		Surface Area per Metre	Second Moment Of Area		Radius Of Gyration		Elastic Modulus		Plastic Modulus		Buckling Parameter	Torsional Index	Warping Constant	Torsional Constant	Designation Size	
			Axis x-x	Axis y-y	Axis x-x	Axis y-y	Axis x-x	Axis y-y	Axis x-x	Axis y-y	u	x	H	J		
lb/ft	kg/m	m²	cm⁴	cm⁴	cm	cm	cm³	cm³	cm³	cm³			dm⁶	cm⁴	in	
118	175.60	2.78	246000	7800	33.1	5.90	5893	535	6808	842	0.856	46.5	13.0	221	33 x 11 1/2	W33
130	193.46	2.79	279200	9067	33.6	6.06	6642	620	7640	974	0.862	41.6	15.2	306		
141	209.83	2.80	310200	10250	34.0	6.18	7334	700	8416	1097	0.867	37.9	17.3	404		
152	226.20	2.81	339700	11360	34.3	6.27	7985	773	9155	1212	0.870	35.0	19.3	514		
169	251.50	2.82	386500	12920	34.8	6.36	8999	884	10310	1383	0.875	30.9	22.1	735		
187	278.29	2.84	430400	14470	34.8	6.38	9950	984	11450	1547	0.874	28.3	25.0	983		
204	303.58	2.85	473000	16050	35.0	6.45	10860	1086	12530	1710	0.874	26.1	27.9	1265		
219	325.91	2.87	513500	17540	35.1	6.49	11710	1180	13550	1864	0.874	24.4	30.7	1569		
243	361.62	2.89	573400	19800	35.3	6.56	12960	1321	15060	2095	0.874	22.3	35.0	2095		
271	403.29	2.91	648200	22660	35.5	6.64	14480	1500	16910	2385	0.875	20.2	40.5	2884		
301	447.94	2.94	729200	25820	35.7	6.72	16120	1692	18920	2704	0.874	18.4	46.7	3910		
332	494.07	2.97	810000	28960	35.9	6.79	17700	1880	20890	3018	0.873	17.0	53.1	5088		
361	537.23	2.99	892800	32280	36.1	6.87	19300	2078	22880	3346	0.873	15.8	59.8	6494		
201	299.12	3.24	479800	31180	35.5	9.05	11220	1559	12640	2405	0.881	31.2	53.2	855	33 x 15 3/4	
221	328.88	3.26	534700	34970	35.7	9.13	12410	1742	14020	2690	0.882	28.4	60.1	1146		
241	358.65	3.27	590000	38800	35.9	9.21	13590	1927	15380	2977	0.884	26.1	67.3	1491		
263	391.39	3.29	659000	43070	36.3	9.29	15030	2146	17020	3314	0.888	23.6	75.5	2008		
291	433.05	3.31	735400	48390	36.5	9.36	16620	2396	18910	3709	0.887	21.6	85.6	2687		
318	473.24	3.33	812500	53680	36.7	9.43	18200	2644	20780	4101	0.888	19.9	95.8	3488		
354	526.81	3.35	913500	60680	36.9	9.50	20330	2968	23230	4616	0.887	18.1	110	4737		
387	575.92	3.38	1012000	67480	37.1	9.59	22170	3280	25570	5113	0.887	16.7	123	6135		
424	630.98	3.40	1120000	74810	37.3	9.64	24260	3614	28140	5652	0.886	15.5	138	7983		
468	696.46	3.43	1253000	84520	37.6	9.75	26810	4044	31270	6342	0.886	14.2	158	10569		
135	200.90	2.96	325300	9424	35.7	6.07	7204	621	8351	982	0.854	46.8	18.4	291	36 x 12	W36
150	223.22	2.97	376400	11240	36.3	6.27	8269	739	9535	1163	0.861	41.3	22.1	422		
160	238.11	2.98	406100	12270	36.6	6.36	8879	805	10230	1266	0.864	38.6	24.2	515		
170	252.99	2.99	436300	13300	36.8	6.42	9501	871	10940	1371	0.866	36.2	26.4	626		
182	270.85	3.00	470700	14460	36.9	6.47	10200	943	11770	1487	0.866	33.9	28.8	767		
194	288.70	3.01	504200	15600	37.0	6.51	10880	1014	12570	1601	0.867	31.9	31.2	926		
210	312.51	3.03	549000	17130	37.1	6.55	11780	1107	13650	1754	0.866	29.7	34.5	1164		
232	345.25	3.04	625200	19480	37.7	6.66	13260	1266	15340	2000	0.872	26.4	39.7	1648		
256	380.97	3.06	696900	21980	37.9	6.72	14660	1416	17030	2247	0.872	24.1	45.2	2199		
286	425.61	3.09	777000	24640	38.1	6.78	16170	1575	18870	2510	0.871	22.2	51.3	2891		
318	473.24	3.11	884100	28580	38.3	6.89	18210	1810	21350	2895	0.872	19.9	60.0	4091		
350	520.86	3.14	981300	32080	38.5	6.95	20010	2013	23590	3235	0.871	18.4	68.1	5337		
387	575.92	3.17	110100	36460	38.8	7.05	22180	2266	26260	3654	0.871	16.8	78.5	7127		
230	342.28	3.42	625800	39160	37.8	9.46	13730	1871	15480	2890	0.883	30.1	75.8	1193	36 x 16 1/2	
245	364.60	3.43	671700	42240	38.0	9.53	14660	2014	16550	3111	0.884	28.3	82.2	1442		
260	386.92	3.44	719600	45440	38.2	9.59	15630	2161	17670	3341	0.885	26.7	88.9	1734		
280	416.69	3.45	785700	49870	38.4	9.68	16940	2366	19170	3658	0.887	24.8	98.3	2188		
300	446.45	3.47	846900	53940	38.6	9.73	18150	2550	20600	3948	0.887	23.3	107	2671		
328	488.12	3.48	936400	59170	38.8	9.76	19880	2801	22640	4344	0.887	21.4	119	3500		
359	534.25	3.50	1031000	65450	38.9	9.81	21700	3081	24820	4790	0.886	19.8	132	4508		
393	584.85	3.53	1144000	73000	39.2	9.89	23840	3415	27390	5320	0.887	18.2	149	5890		
439	653.30	3.56	1292000	83050	39.4	9.99	26590	3854	30730	6014	0.887	16.5	171	8124		
527	784.26	3.62	1593000	103300	40.0	10.2	31980	4728	37340	7439	0.887	14.0	218	13730		
650	967.30	3.70	2033000	133900	40.6	10.4	39540	6003	46810	9504	0.886	11.8	292	24930		

Note: Calculated based on 1 kg = 2.2046 lb and 1 m = 3.2808 feet.
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